

Prasanna Agarwal

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EDUCATION

Technical University Dortmund, *Master of Science in Automation and Robotics*

present | Dortmund, Germany

Relevant Coursework: Modelling and Control of Robotic Manipulator, Control Theory and Application, Computer Systems, Mobile Robots, Network Mobile Robot Systems, Process Control, Online Problems, Automated Driving, 3D Computer Vision, Intensive Course in Statistics for Researchers in Engineering Sciences, Application of Robots.

SRM Institute of Science and Technology, *Bachelor of Technology in Mechatronics Engineering*

May 2022 | Chennai, India

GPA: 8.97/10 (10 being the best possible grade)

PROJECTS

Trajectory planning in mobile robots with Diffusion Probabilistic models, *Masters Thesis*

June 2025 – Present

- Developed a hierarchical diffusion-based trajectory planning framework trained on large scale BiRRT datasets in SLAM-based environments.
- Integrated trained models into the TurtleBot navigation stack using ROS2 and Nav2 to demonstrate autonomous navigation.
- Combined generative modeling with classical motion planning to improve path feasibility and robustness.

Voice Controlled TurtleBot3 Navigation

- Developed a ROS 2 package to control TurtleBot3 in Gazebo via spoken commands, publishing velocity commands over /cmd_vel.
- Built multi-accent speech recognition (5 English/German locales) with fuzzy matching for robust command interpretation.
- Designed a command-queueing system supporting multi-step voice sequences with automatic timed execution & safety auto-stop

AI-Based SI/PI/EMC-Compliant PCB, *Signal Integrity Subgroup*

October 2023 – June 2024

Enhanced interference-resistant PCB design by applying AI/ML methods, using CNN-based image processing and feature detection to identify SI/PI/EMC-related patterns in PCB layouts. Combined microelectronics expertise with machine learning to develop robust, interference-resistant microelectronic systems.

Contour Crafting with IoT-based End-effector

Developed a downscaled 3D printing system for constructing walls, with IoT-controlled end effector to manage concrete flow and enable instant stoppage in case of issues.

Robotics Mini projects

- Built embedded robots using IR/LDR sensors, motor drivers, wireless modules, and embedded C: Obstacle Avoider, Tabletop-edge Detector, PC-Controlled Robot, Light-Sensitivity Robot, Wireless Robot, Pickup-and-Place Robot, and Line Follower.
- Delivered autonomous navigation, sensor-based control, and wireless motion handling across all builds.

EXPERIENCE

Johnson Electric, *Student Lab Assistant*

June 2023 – October 2023 | Halver, Germany

Conducted extensive tests on Micro-switches:

- Lifetime Testing/Force Measurement: Performed multiple tests to evaluate durability and detect early wear patterns.
- Contact Resistance/Isolation Testing: Conducted various tests to ensure compliance with safety and reliability standards.
- Electrical/Mechanical Lifetime: Analyzed performance under load and high-cycle conditions to predict failure modes.

Achievements:

- Enhanced Reliability: Improved product performance and durability by identifying early failure trends through structured testing.
- Informed Design Improvements: Delivered data-driven insights that guided material, tolerance, and structural optimizations, shortening redesign cycles and reducing field failures

TATA Fico, *Research Intern*

January 2021 – March 2021 | Pune, India

Authored Paper on "Encoding and Decoding of Lithium-Ion Batteries":

- Task: Researched and wrote a detailed paper on techniques for lithium-ion battery performance data.
- Output: Developed comprehensive insights and methodologies for battery management system.

Gains:

- Enhanced Knowledge: Deepened understanding of battery data processing, ASIC, SOA, BMS, and FET.
- Contributed to Field: Provided insights supporting safer BMS development and advancing lithium-ion battery technology for EV applications.

SKILLS

- **Robotics frameworks & Tools:** ROS2, Gazebo, RViz, SLAM, Motion Planning (BiRRT, RRT, A-Star, RRT-Star), Nav2, Reinforcement Learning, Localization, Pose Estimation, Kinematics & Dynamics, Teleoperation
- **Programming languages/Libraries:** Python (NumPy, PyTorch, Scikit-learn, OpenCV, TensorFlow, Keras, CUDA), C, Deep Learning, NLP, Image Processing, Speech Recognition
- **Others:** Git, MLOps, Linux, MATLAB, SolidWorks, Trajectory Optimization, Sensor Integration, Technical Writing

CERTIFICATES

Computer Vision Basics: Coursera, **Kaizen Robotics Program:** IIT-Madras, **Bosch Rexroth hydraulics industrial training:** Electric and Basic, **T-Zero 3.0 (Arduino Programming):** SRM-IST

PUBLICATIONS

Contour crafting based 3D printer

Encoding and Decoding of Li-ion Voltage reading

LANGUAGES

English: Professional proficiency • **German:** B1 Proficiency • **Hindi:** Native Proficiency

CO-CURRICULAR

- Conducted numerous Robotics workshops for kids (Age group: 8-14) on hydraulic JCB arms.
- BOT Quest & Codathon Competition: SRM-IST